

Robert L. Zacchigna

[About - Robert-Zacchigna.github.io](https://robert-zacchigna.github.io) | [Contact - Robert-Zacchigna.github.io](https://robert-zacchigna.github.io) | linkedin.com/in/robert-zacchigna

Seasoned engineer experienced with designing/implementing machine learning, data pipelines and analytics solutions in large, enterprise, environments. Proficient in DevOps, PySpark, IaC, Git, SQL, Containerization, and Orchestration with a focus on delivering data-driven insights to support enterprise teams and interpolating trends initiatives.

PROFESSIONAL EXPERIENCE

Capital One

May 2022 – Present

Senior Assoc Software Engineer

Chicago, IL

- Designed technical documentation for and wrote internal python libraries/utilities to help facilitate data orchestration using an AWS Fargate deployed Apache Airflow platform, resulting in a +35% reduction in duplicated/unneeded code, +20% pipeline development efficiency and a vastly more maintainable platform.
- Rebuilt legacy Python application to utilize event-driven architecture (SNS, SQS, PySpark and Glue), in AWS, resulting in a far more scalable and maintainable application and a +15% decreased resource usage/cost among +200 users.
- Facilitated the solution architecture (IaC - Terraform) of new external partner data pipelines (Glue, PySpark, Fargate, SNS, SQS) using a micro-service architecture, as the designated SME within my team (including creating technical documentation). This architecture proved to be ~15% more resource efficient than other proposed solutions.

Discover Financial Services

June 2019 – May 2022

Data Engineer

Riverwoods, IL

- Troubleshooted, architected and implemented improved enterprise financial data pipelines leveraging PySpark, SQL, and Kafka resulting in a +20% performance increase in loading high-velocity financial data into Snowflake.
- Optimized deployed financial data models utilizing PySpark, Jenkins, and in-house modeling tools to on-premises servers, resulting in +15% lower resource cost/usage while adhering to stringent data governance and security standards.
- Collaborated, designed and implemented a containerized (OpenShift) multi-platform development application with cross-functional teams to address current internal development tool short-comings, resulting in +30% increase in development efficiency across +800 internal users.

EDUCATION

Bellevue University, Master of Science – Data Science

- Personal website containing my projects: <https://robert-zacchigna.github.io/>

Illinois State University, Bachelor of Information Technology – Systems Analyst/Development

Honors: *Cum laude*, National Society of Collegiate Scholars, Dean's List

TECHNICAL SKILLS

- Data Science:** Modeling, Machine Learning, Tensorflow, FastAPI, PyTorch, Pandas, PySpark, Spark, NumPy
- Data Engineering:** Analytics, Databricks, ETL, Airflow, Glue, Kafka, Orchestration, Data Warehouse/Lake/Processing
- DevOps:** Docker, Kubernetes, Git, GitOps, Agile, Jenkins, CI/CD, Unit Testing, MLOps, Artifactory, Linux
- Cloud:** AWS, Terraform, Fargate, EC2, ECS, EMR, Glue, RDS, S3, EFS, SNS, IAM, Lambda
- Database:** NoSQL, MySQL, Postgres, SQLite, MongoDB, Aurora, DynamoDB, Snowflake
- Programming:** Python, Scala/Java, SQL, Bash, Typescript/Javascript, REST API, PowerShell
- Visualization:** Tableau, Plotly, Power BI, matplotlib, seaborn

CERTIFICATIONS AND PROJECTS

- [AWS Certified Machine Learning – Specialty](#)
- [AWS Certified Solutions Architect – Associate](#)
- Python Etrade API Module ([PyEtrade](#))**
 - Significantly contributed to an open source python Etrade API library enhancing code quality, unit-tests and cleanliness.
 - Realigned missing functions and parameters to be in line with API documentation specifications and adding several missing/undocumented API endpoints to the module.
 - Received praise and recognition from the original author for my significant code contributions (+200 Stars).
- Malaria Cell Image Classification ([FastAI and Keras/Tensorflow Comparison](#))**
 - Cross compared FastAI and Keras neural network models, for determining if an image (~7000 images) of a cell is infected with malaria or not, to see the differences between how the models are developed and performed.
 - Both models scored very well, with a +94% prediction accuracy.
- Multi-Processing Media File Management CLI ([MKVAudioSubsDefaulter](#))**
 - Created a Python CLI to more easily edit/organize tracks and manage metadata of Matroška (MKV) media files without having to remux (re-encode) the files.
 - Multi-processing allows this to be easily scalable and automatable to a large amount of files, processing over 1500 files in less than 4 mins (+375 files/s).